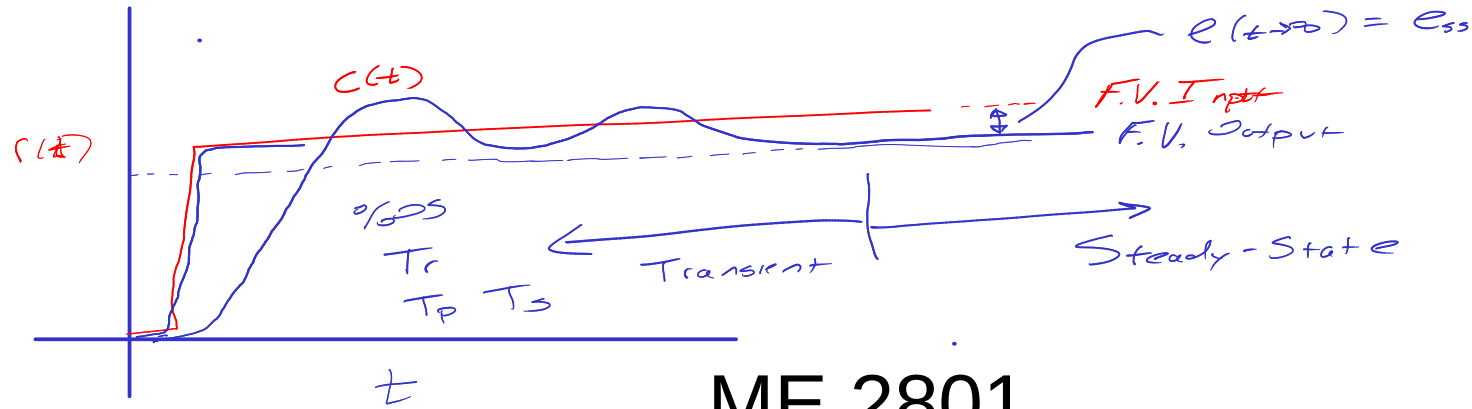


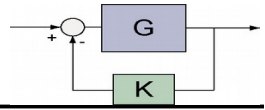


ME 2801

Introduction to Control Systems

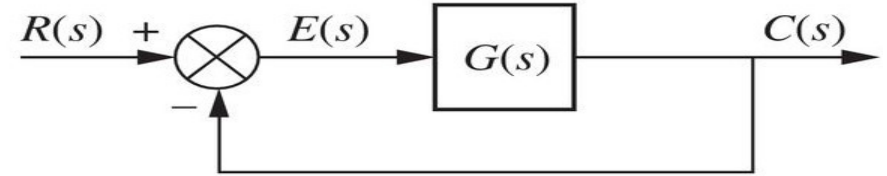
W5: Steady State Error
and System Type

Steady-State Error



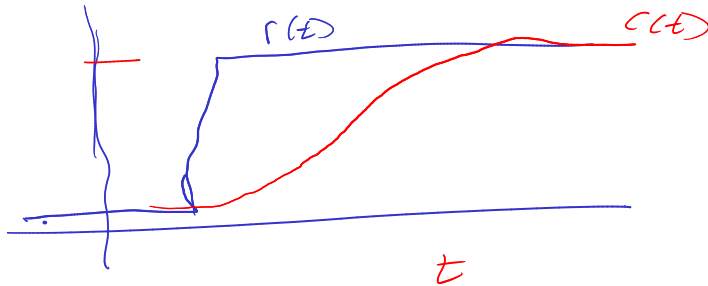
So what? Closed-loop system response specifications

- Transient response
- Stability
- Steady-state errors

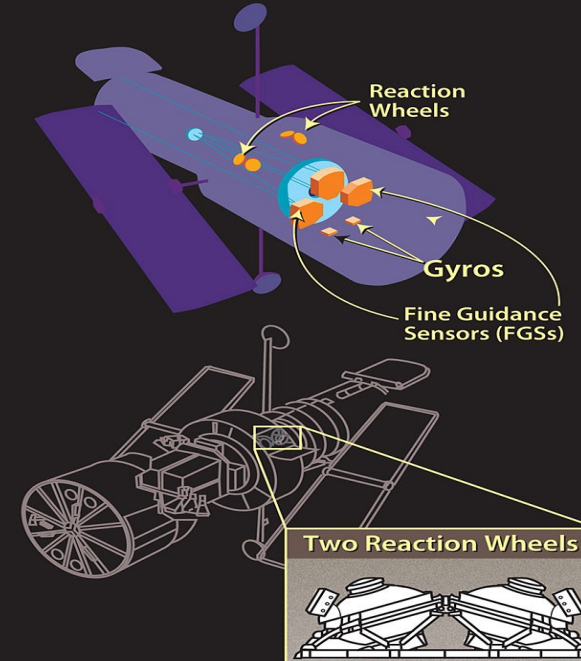
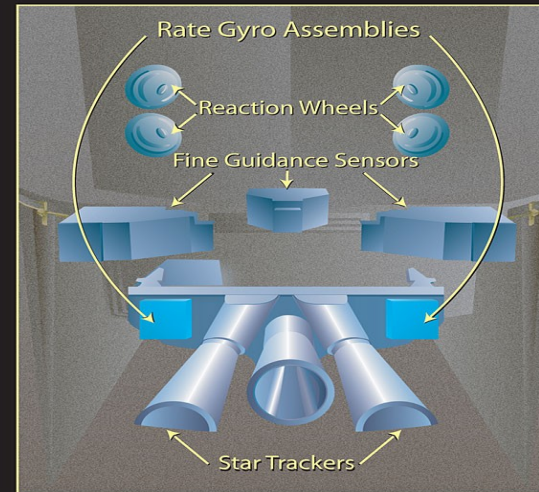


Input Type

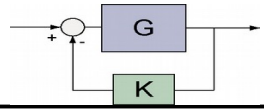
1. Step (position)
2. Ramp (velocity)
3. Parabola (acceleration)



Hubble's Pointing Control System



Steady-State Error

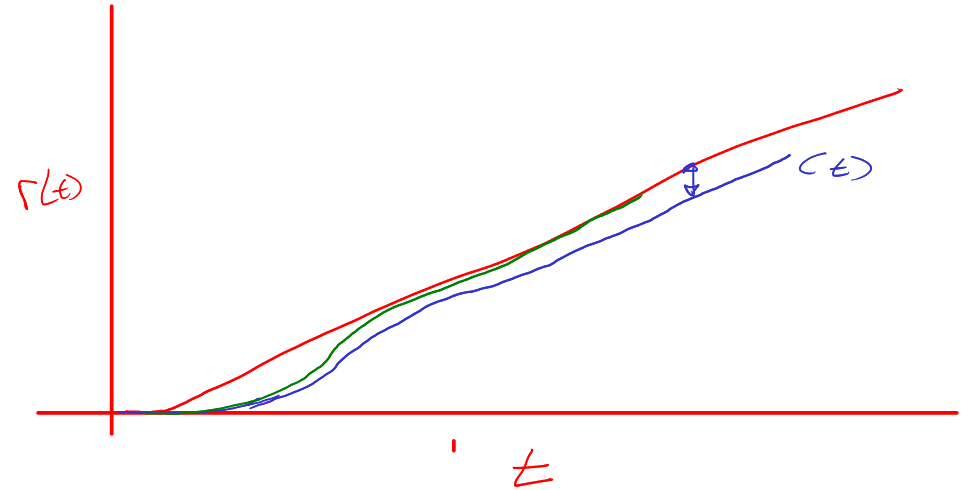
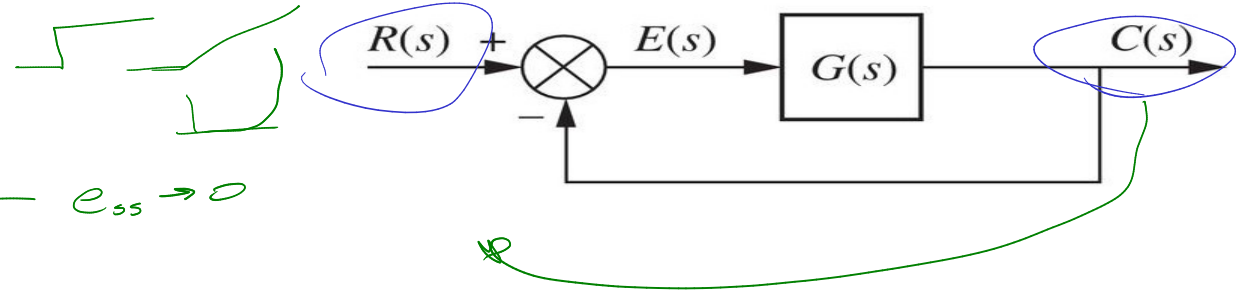


So what? Closed-loop system response specifications

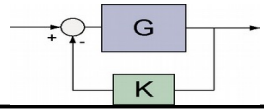
- Transient response
- Stability
- Steady-state errors

Input Type

1. Step (position) ✗
2. Ramp (velocity)]
3. Parabola (acceleration)]

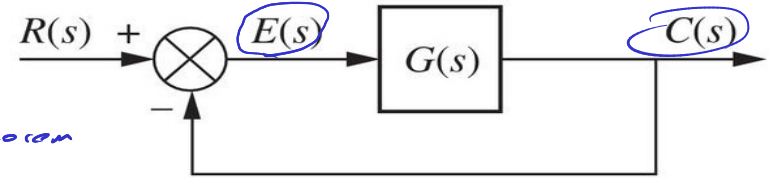


Steady-State Error



Steady-state error as performance specification for feedback system.

- Given a unity feedback system, what is the steady-state response?



Final Value Theorem

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

$$E = R - C \quad C = G E$$

$$E = R - G E$$

$$(1 + G) E = R \rightarrow \frac{E}{R} = \frac{1}{1 + G}$$

S.S.E. Sensitivity

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} s \frac{R}{1 + G}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}$$

- For the following test inputs

- Step (position) $R(s) = \frac{1}{s}$

$$r(t) = y(t) = 1$$

$$e(\infty) = e_{\text{step}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s)}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)}$$

- Ramp (velocity)

$$R(s) = \frac{1}{s^2}$$

$$r(t) = t$$

$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)}$$

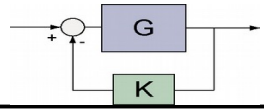
- Parabola (acceleration)

$$r(t) = \frac{1}{2} t^2$$

$$R(s) = \frac{1}{s^3}$$

$$e(\infty) = e_{\text{parabola}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}$$

USV Example



Consider steady-state error for..

- Surge proportional control
 - Step and Ramp input

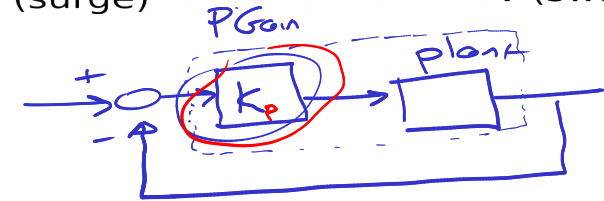
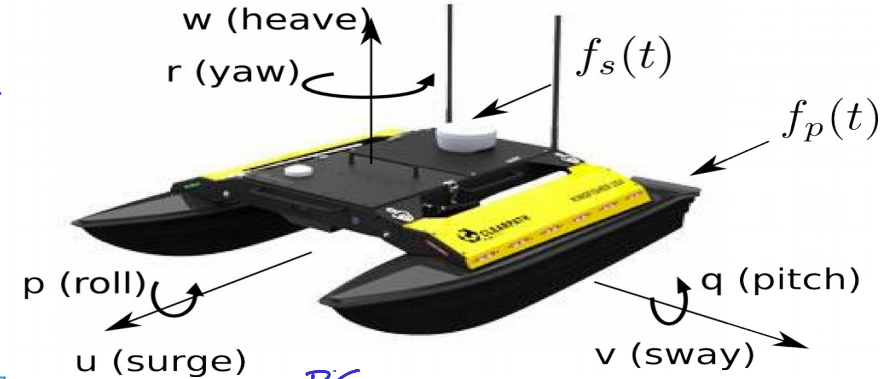
Unity Feedback Sys

$$G_u(s) = \frac{U(s)}{T(s)} = \frac{1}{Ms + D}$$

- Yaw proportional control
 - Step and Ramp input

$G_r(s) = \frac{R(s)}{\tau(s)} = \frac{1}{s(Is + d)}$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$



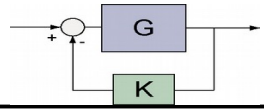
	Surge Control	Yaw Control
Step Input $R(s) = \frac{1}{s}$	① $Const = \frac{1}{1 + K/D}$	③ $\emptyset$
Ramp Input $R(s) = \frac{1}{s^2}$	② ∞	④ $Const = \frac{1}{K/D}$

④ $e_{ss} = \lim_{s \rightarrow 0} \frac{1}{sG(s)} = \frac{1}{s \frac{K_p}{s(Is + d)}} = \frac{1}{K/D}$

② " $e_{ss} = \lim_{s \rightarrow 0} \frac{1}{sG(s)} = \frac{1}{1 + K/D}$

③ $e_{ss} = \lim_{s \rightarrow 0} \frac{1}{(1 + G(s))} = \frac{1}{1 + K_p \left(\frac{1}{s(Is + d)} \right)} = \frac{1}{1 + \infty} = \lim_{s \rightarrow 0} \frac{1}{s \frac{1}{Ms + D}} \Rightarrow \infty$

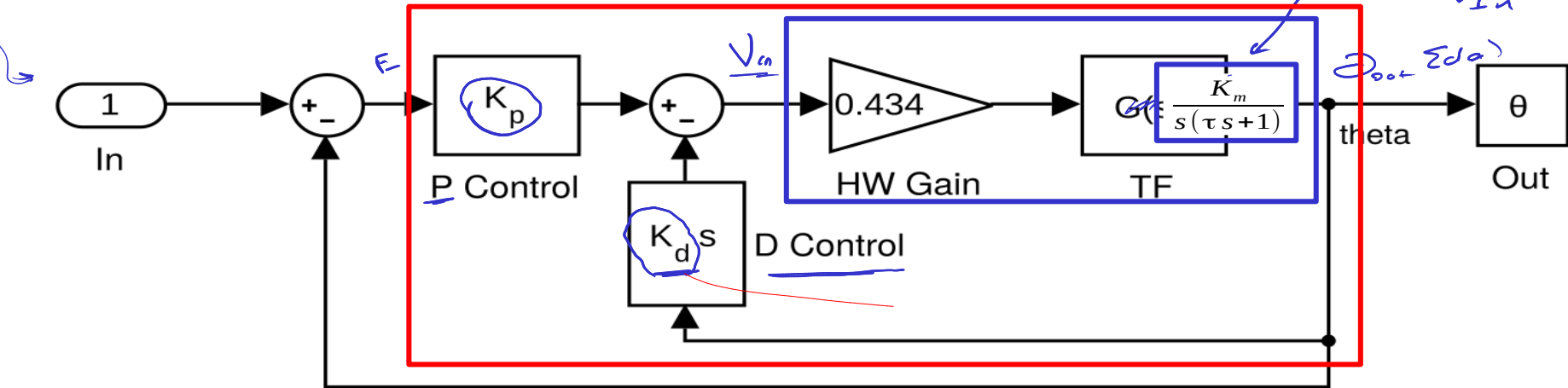
Example



Given ECP **open-loop position** response, what is the **closed-loop** steady state error for...

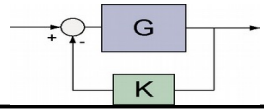
- Step, position command
- Ramp, velocity command
- Parabola, acceleration command

PD = Proportion + Derivative



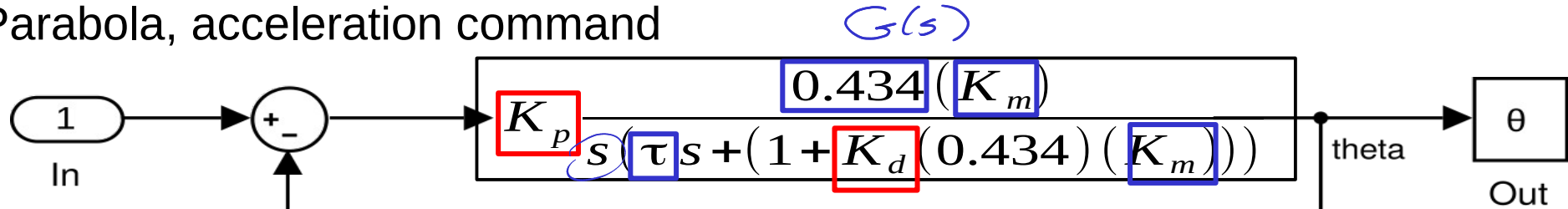
$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}$$

Example



Given ECP **open-loop position** response, what is the **closed-loop** steady state error for...

- Step, position command
- Ramp, velocity command
- Parabola, acceleration command



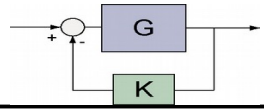
①
$$e(\infty) = e_{\text{step}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s)}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)}$$

①
$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)}$$

Const

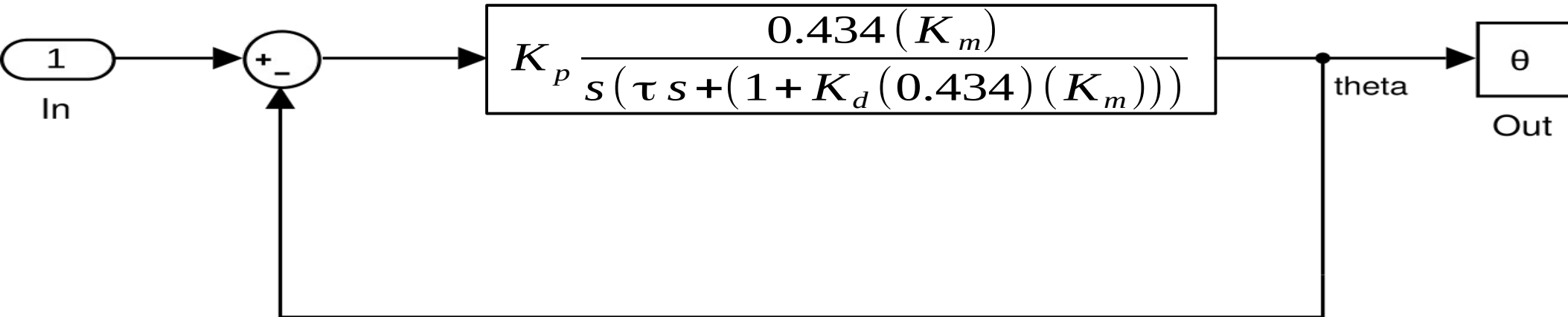
②
$$e(\infty) = e_{\text{parabola}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2G(s)} \rightarrow \infty$$

Example

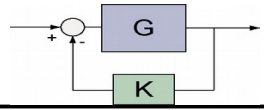


Given ECP **open-loop position** response,
what is the **closed-loop** steady state error for...

- Step, position command $\longrightarrow e_{step}(\infty) = 0$
- Ramp, velocity command $\longrightarrow e_{ramp}(\infty) = \frac{1 + K_d(0.434)(K_m)}{K_p(0.434)(K_m)}$
- Parabola, acceleration command $\longrightarrow e_{parabola}(\infty) = \infty$



System Type: Definition



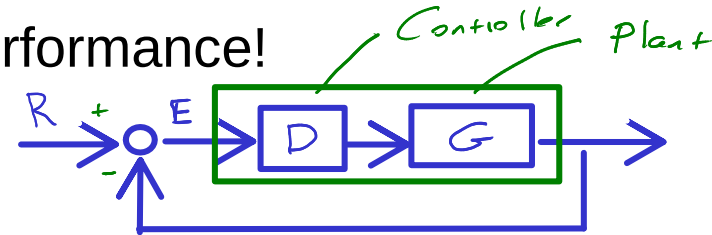
System Type

- degree of the polynomial for which the steady-state error is a nonzero finite constant
- equals the number of "pure integrators" in the open-loop transfer function.
- We can analyze the open-loop dynamics predicting the closed-loop tracking performance!

$\rightarrow$ Input $\left\{ \begin{array}{l} \text{Step} = j\omega(t) \\ \phi \end{array} \right. \quad \left| \quad \begin{array}{l} \text{Ramp} = t \\ 1 \end{array} \right. \quad \left| \quad \begin{array}{l} \text{Parabola} = t^2 \\ 2 \end{array} \right.$
 $\text{poles @ } \phi$

$$E = \left(\frac{1}{1 + \boxed{DG}} \right) R$$

$L(s)$

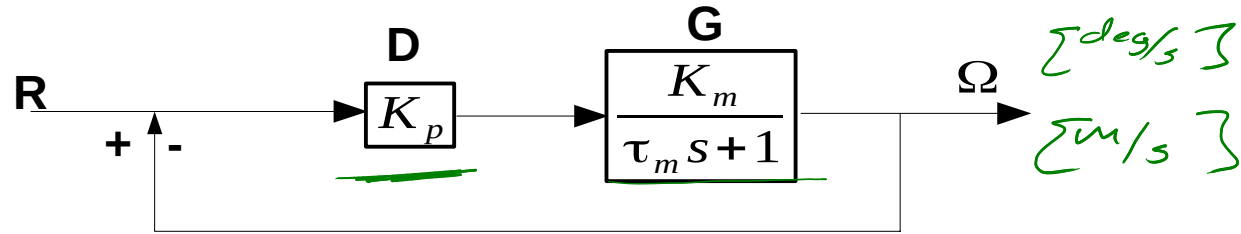
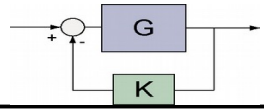


"Sensitivity" TF

So What?

- Since $D(s)$ is our controller, we can design the controller to change the system type.
- System type can tell us, at a high level, what type of control to accomplish our goals.

Example: Speed P Control



Find steady-state error

$$E = \frac{1}{1 + DG} R$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

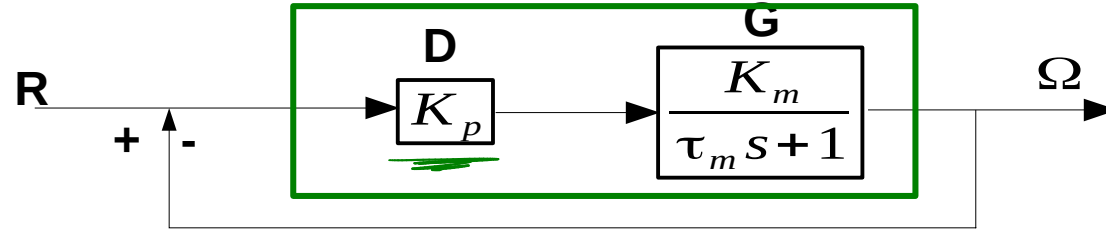
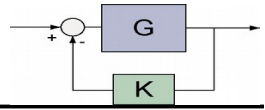
Final-value property

$$e_{ss} = \lim_{t \rightarrow \infty} e_{ss}(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \left(\frac{s}{1 + DG} \right) R$$

Now examine some test inputs: **step**, ramp, parabola

- Polynomial degree: step=0, ramp=1, parabola=2

Example: Speed P Control



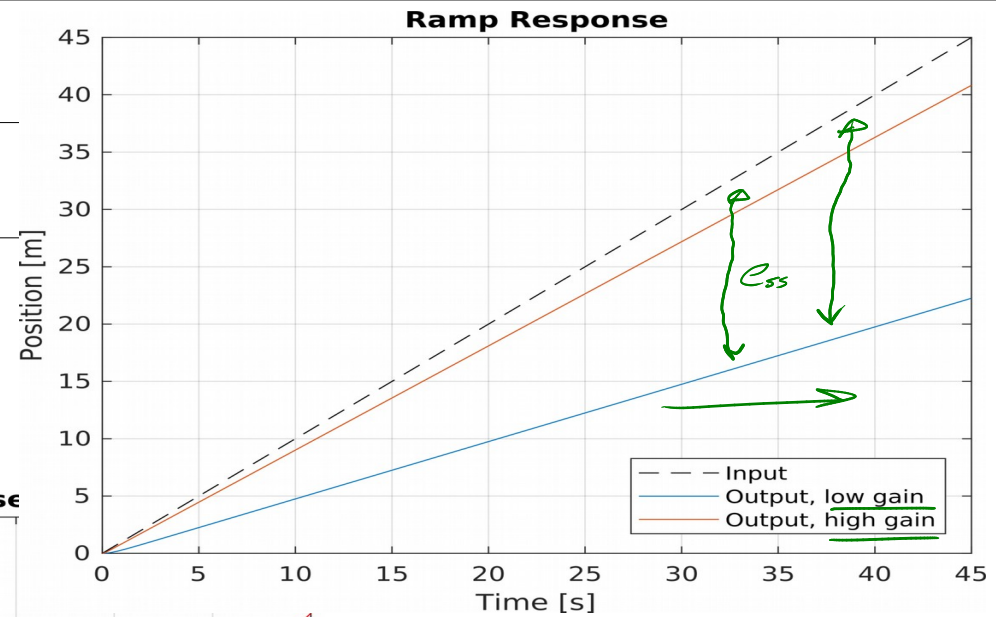
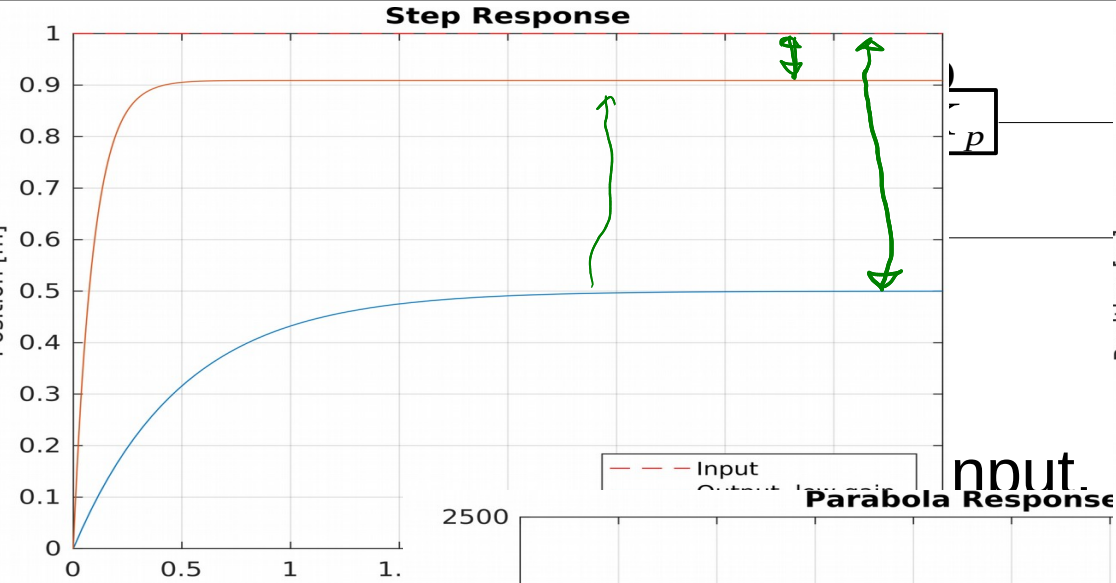
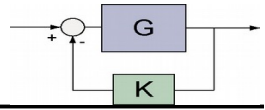
Proportional Control:

For speed output and step input,
We will always have constant,
non-zero steady-state error!

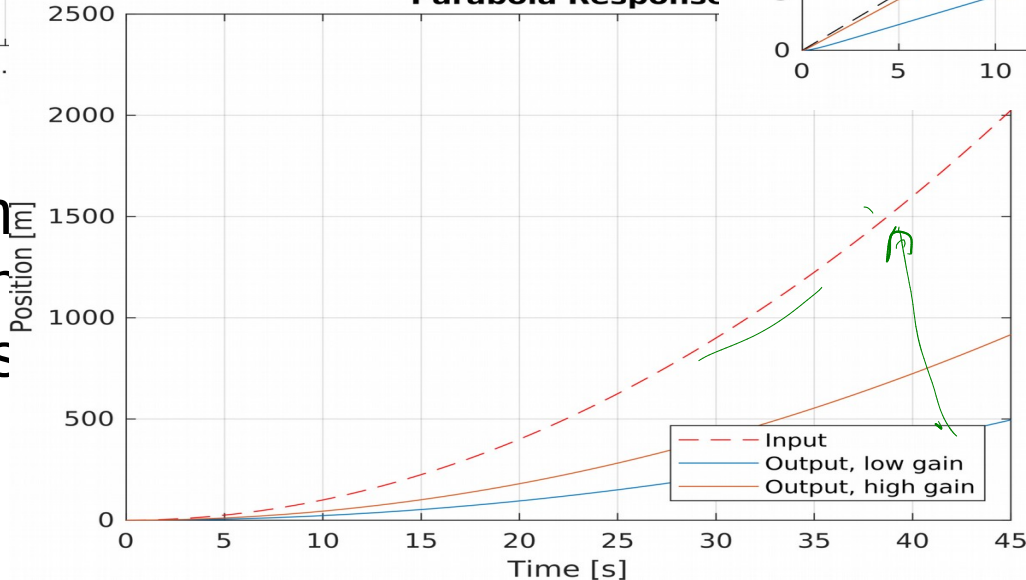
$$C_{ss} = \lim_{s \rightarrow 0} \frac{1}{1 + \frac{K_p K_m}{\tau_m s + 1}} = \frac{1}{1 + K_p K_m}$$

The system ($DG = L$) is type 0, meaning...
a step input (order=0) will result in constant
steady-state error.

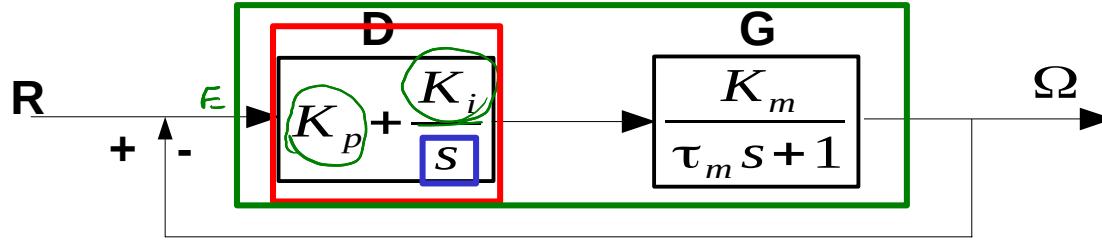
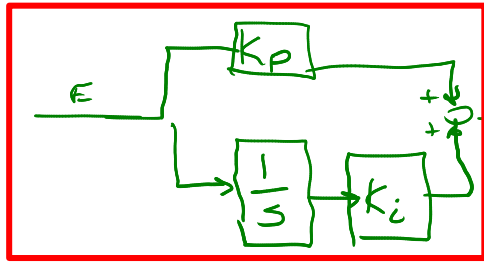
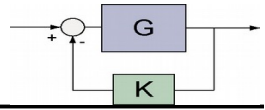
Example: Speed P Control



The system
a step in
steady-s



Example: Speed PI Control



So let's change the system type
and get rid of the steady-state error

- Add an integrator...

$$e_{step} = 0$$

$$e_{ramp} = \text{const} \cdot 1$$

$$e_{parab} = \infty$$

$$\underline{DG = L} = \frac{K_m (\underline{K_p} s + \underline{K_i})}{s (\tau_m s + 1)}$$

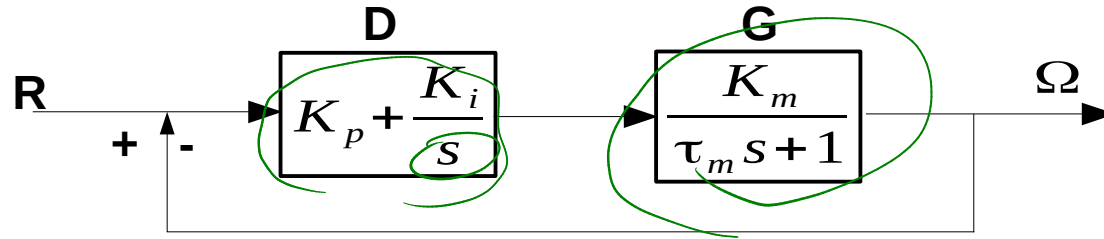
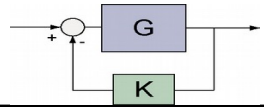
↑
pure integrator
pole @ zero

Again, examine some test inputs:

step, ramp, parabola

- Polynomial degree: step=0, ramp=1, parabola=2

Example: Speed PI Control



Proportional - Integral Control:

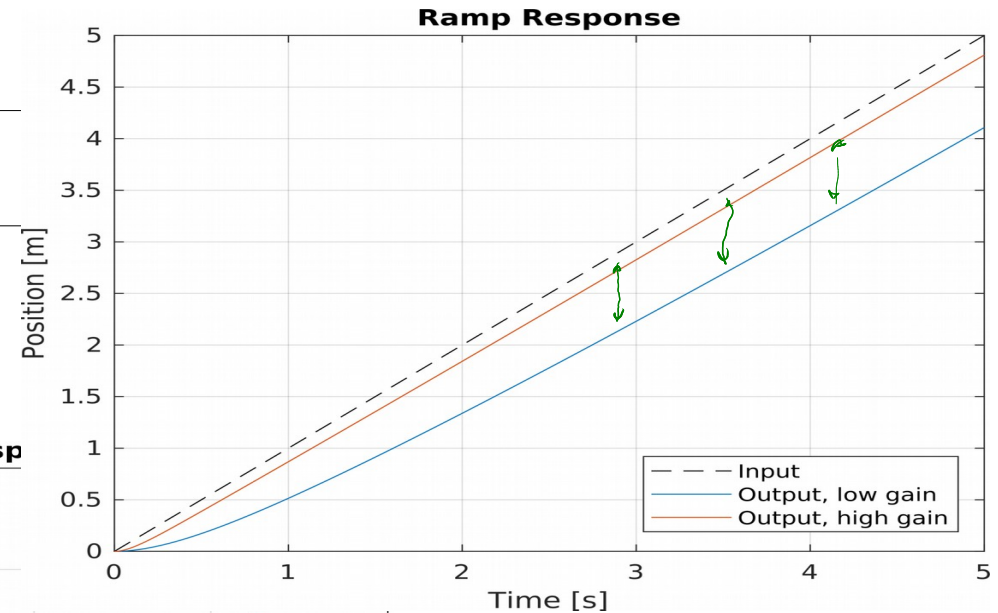
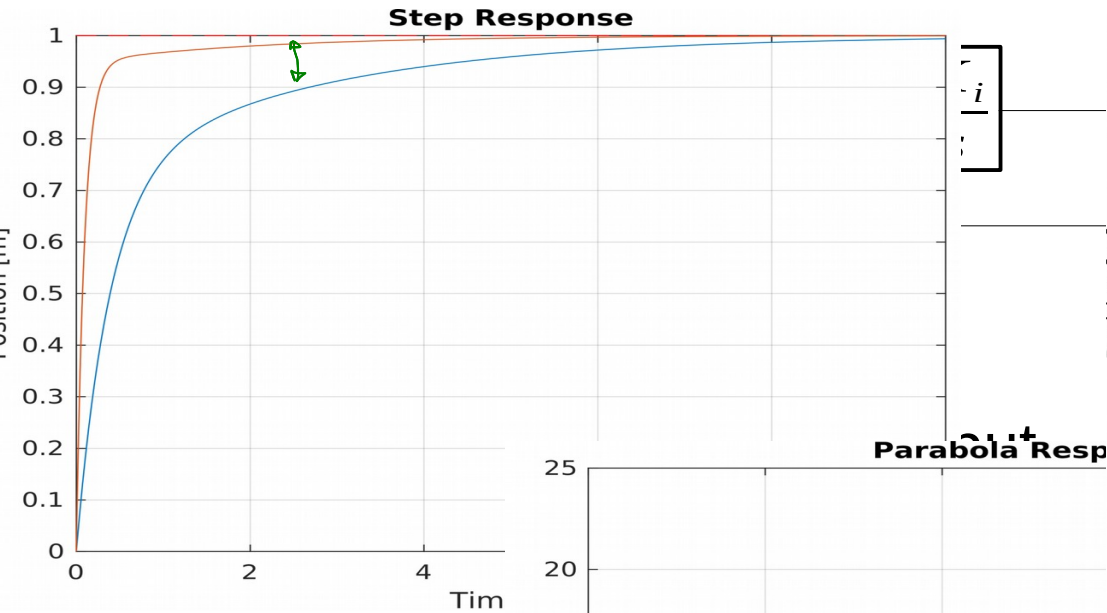
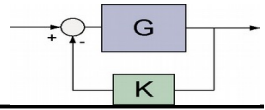
For speed output and step input,

We will always have zero steady-state error

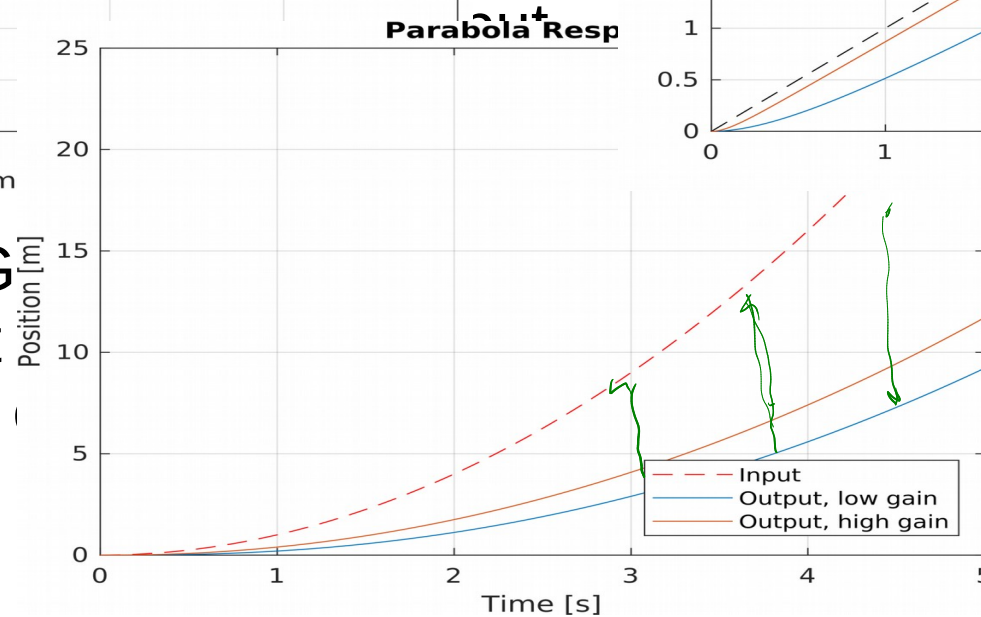
The system ($DG = L$) is type 1, meaning...

a ramp input (order=1) will result in constant steady-state error

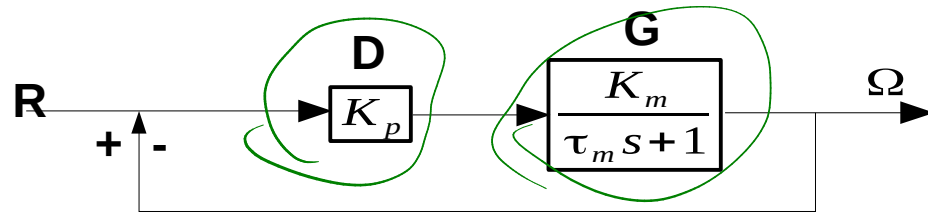
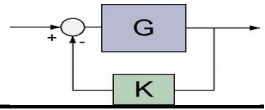
Example: Speed PI Control



The system (DG)
a ramp input
steady-state

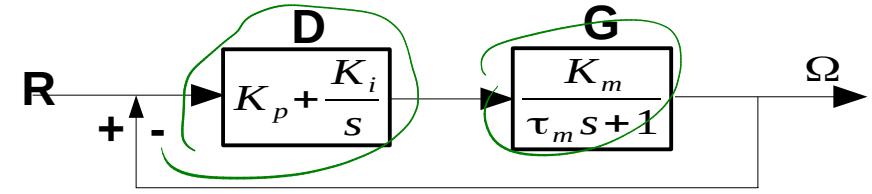
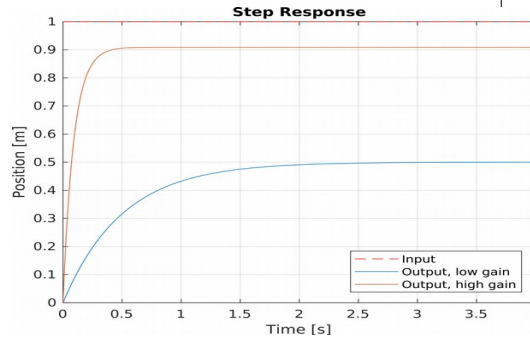


Speed Control Summary



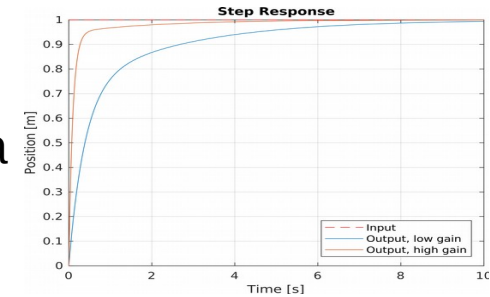
P Control

- Open-Loop System is Type 0
 - No pure integrators.
- Steady-State Error
 - Constant for step
 - Infinite for ramp
 - Infinite for parabola



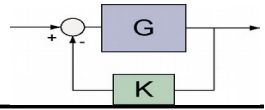
PI Control

- Open-Loop System is Type 1
 - One pure integrators.
- Steady-State Error
 - Zero for step
 - Constant for ramp
 - Infinite for parabola



Adding an integrator changes the dynamics!

Summary



Summary of system type, inputs and constants

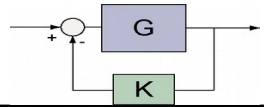
TABLE 7.2 Relationships between input, system type, static error constants, and steady-state errors

Input	Steady-state error formula	Type 0	Type 1	Type 2
		Static error constant	Static error constant	Static error constant
Step, $u(t)$	$\frac{1}{1 + K_p}$	$K_p = \text{Constant}$	$K_p = \infty$	$K_p = \infty$
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	$K_v = \text{Constant}$	$K_v = \infty$
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	$K_a = 0$	$K_a = \text{Constant}$

$$e_{ss}(\text{step}) = \lim_{s \rightarrow 0} \frac{1}{1 + G(s)} \rightarrow \lim_{s \rightarrow 0} L(s)$$

$$\frac{32}{s(s+8)(s+10)}$$

$$\frac{42}{s^2(s^2+3s+18)}$$



System Type

- degree of the polynomial for which the **steady-state error** is a nonzero finite constant
- equals the number of “pure integrators” in the open-loop transfer function.
poles @ zero

Design Takeaway

- If you want zero steady-state error,
then you need at least one pole at zero (pure integrator)
in the open-loop TF: $L = \underline{D}\underline{G}$